Paper Code:

Measurement-Oriented Microscopy Reconstruction for Cross-Magnification Micro-Displacement Inspection

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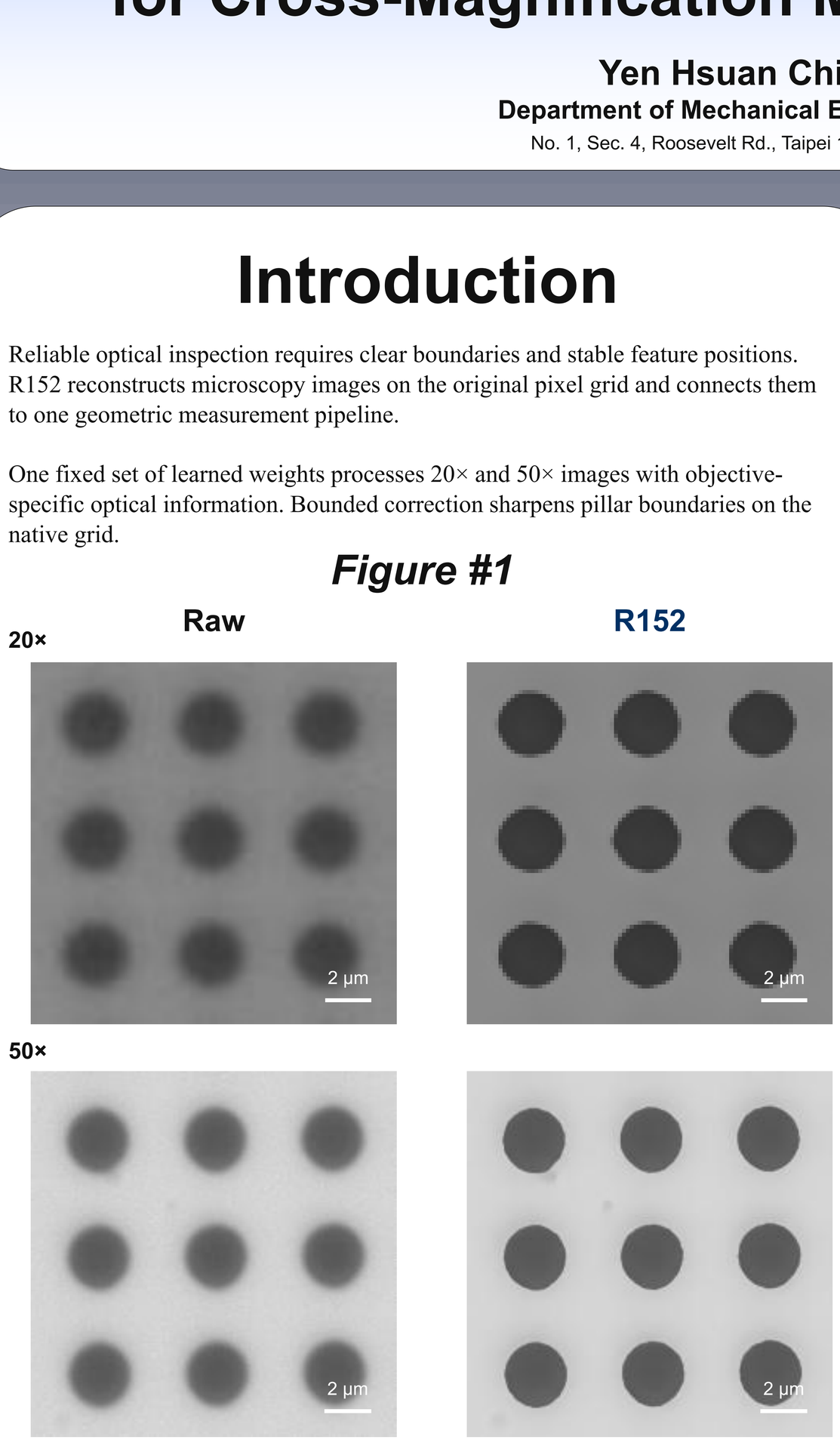
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Introduction

Reliable optical inspection requires clear boundaries and stable feature positions. R152 reconstructs microscopy images on the original pixel grid and connects them to one geometric measurement pipeline.

One fixed set of learned weights processes 20× and 50× images with objective-specific optical information. Bounded correction sharpens pillar boundaries on the native grid.

Figure #1



Representative pillar fields at 20× and 50×. R152 yields clearer boundaries on the native pixel grid. Scale bars: 2 µm.

Methods

R152 uses U-Net context [1], anti-aliased processing [2], PSF guidance and bounded output correction. The optical inputs comprise 25 kernels at 20× and one calibrated kernel at 50×.

All image methods share one LSC pipeline [3, 4]. Each pillar uses 720 radial samples and three center-refinement passes. The median horizontal shift of nine pillars defines each step.

Table #1. Locked acquisition conditions

Objective	Step (µm)	Pairs	µm/pixel
50×	0.5	30	0.0690
50×	1.0	30	0.0690
20×	0.5	30	0.1725
20×	1.0	30	0.1725

Step SD: mean of four condition-level standard deviations.
Nominal MAE: mean absolute difference from the set step.
Within-grid SD: median of 120 nine-pillar standard deviations.

Results

60.8%

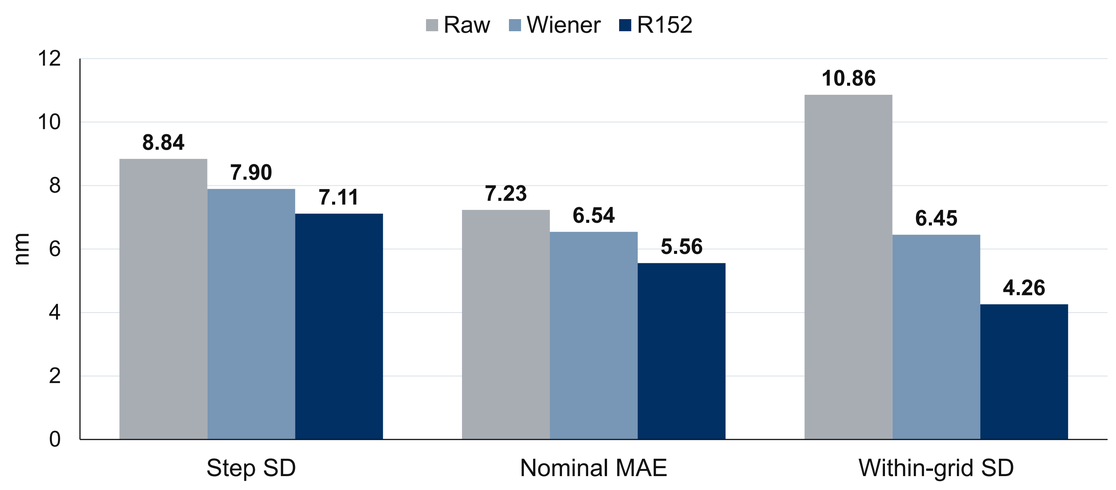
120 paired steps
20× and 50×

Lower median within-grid SD versus Raw

Table #2. Aggregate displacement results (nm)

Method	Step SD	Nominal MAE	Median within-grid SD
Raw	8.84	7.23	10.86
Wiener	7.90	6.54	6.45
R152	**7.11**	**5.56**	**4.26**

Figure #2



R152 achieves the lowest measured values for all three criteria across the four imaging conditions.

Conclusions

R152 reduces median within-grid variation from 10.86 nm to 4.26 nm, a 60.8% reduction relative to Raw.

Step SD and nominal MAE decrease by 19.5% and 23.1%, respectively. The same learned weights deliver the lowest measured values in every tested condition.

Step variation reaches 0.056 pixel at 20× and 0.067 pixel at 50×. R152 combines clear pillar boundaries with stronger positional consistency across both objectives.

Bibliography

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[3] T. F. Chan and L. A. Vese, Active contours without edges, IEEE Trans. Image Process., 10(2), 266–277, 2001.
[4] N. Chernov and C. Lesort, Least squares fitting of circles, J. Math. Imaging Vis., 23, 239–252, 2005.